PSYCH 357

Student Experience Project

In-Class Learning

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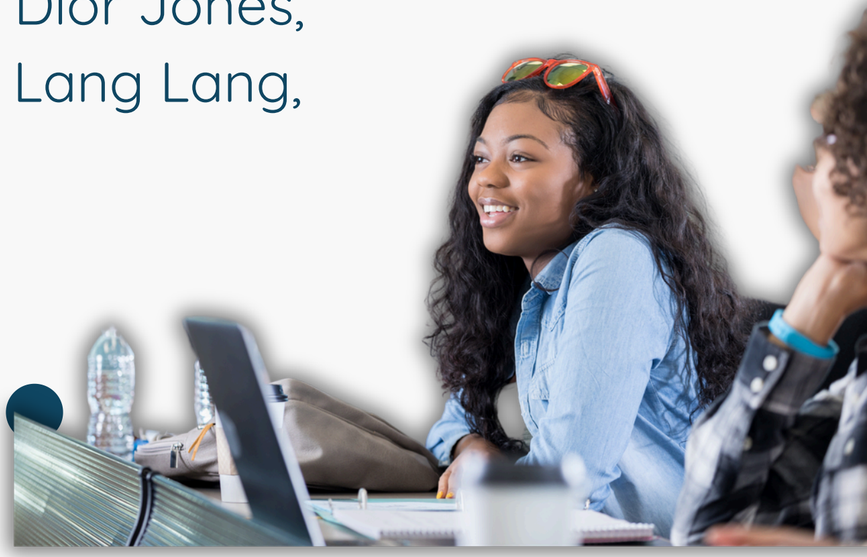
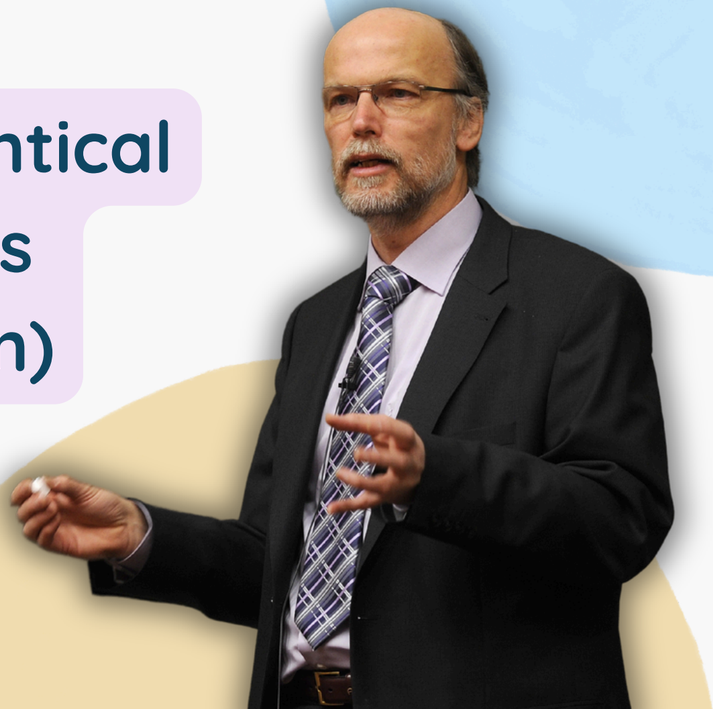


Research Overview - Central Question & Goal

Main Question: Does **gender, race, and age of instructor influence student evaluation** of **teaching quality and engagement?**

Goal: To **examine student perceptions** of **identical lectures** delivered by **different instructors (Young asian woman vs. Older white man)**



Learning Bias

→ When a student's ability to learn or stay engaged is affected by their perception of the instructor's identity, not by the quality or content of the lecture (Culver,2010)

Our Study

→ A systematic influence on students' evaluations based on the instructor's perceived gender, race, or age, rather than actual teaching performance

Literature Review

Instructor Bias

- Sometimes **students judge instructors based on their gender, race, age, or accent rather than how well they teach** (Uttl, et.al 2017).

How it shows up:

- **Female instructors getting lower ratings** for the same performance, **instructors of color being seen as less knowledgeable**
- **Younger or non-native English speaking instructors being viewed as less competent** (Chávez, 2019)

Impact on Evaluations:

- **identity-based assumptions can lead to lower scores in student evaluations**

Why it Matters:

- **reflect bias, not actual teaching quality—hurting instructors in hiring, promotions, or pay decisions**

Gender Bias

- **Studies have found that female instructors often get lower ratings, especially in fields like STEM or when they don't fit traditional expectations (like being overly "nice" or "nurturing").**

Experiment

- **Students rated the "male" instructor higher, even though it was the exact same person, same content, and same teaching style** (MacNell, Driscoll, & Hunt, 2015).
- **This showed that just being perceived as a woman was enough to lower a teacher's scores—purely based on gender bias, not actual performance.**

Race Bias

Asian instructors may be perceived as less relatable or authoritative compared to White counterparts, particularly in **Western academic contexts** (Irizarry, 2019)

Prior studies suggest racialized expectations influence perceived credibility and engagement (Meyer & Zane, 2013)

Age Bias

Older instructors often **perceived as more experienced or knowledgeable** (Semlak, 2008)

Younger instructors—especially women of color—may be **underestimated due to stereotypes or lack of "professorial image"** (Staff, 2021)

Background Info - Cont.

Hypothesis

- We believe that **students will exhibit more effective in-class learning when instructed by male instructors compared to female instructors.**
  - Are students more engaged with one instructor over another?
  - Do students perceive instruction quality differently?

Method and Measures

Independent Variable

Two instructors:

Young Chinese American Female Professor

Old Caucasian American Male Professor

Dependent variable

Student perceived quaality of the lecture

Four indicators:

1. Good instruction ↔ Bad instruction
2. I am learning/engaged ↔ Not learning/not engaged

Operational Definitions

- **Teaching Quality:** Defined as students' **moment-to-moment ratings on a scale from good instruction to bad instruction.**
- **Student Engagement:** Defined as **students' ratings on a scale from engaged/learning to not engaged/not learning**. (Culver, 2010)

These continuous ratings captured real-time perceptions during the lecture.

Our Approach

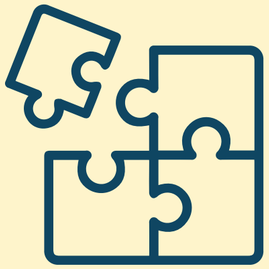
Kitchen Sink Approach:

An approach where many or all available variables are included in an analysis or model. Our research will account for all potentially relevant factors such as gender, age, and race. We cannot conclude which variable cannot conclude which variable will have the strongest influence on the outcome beforehand

Procedure - data collection

Conset form
& Pre-questionaires

- **Demographics (age, gender, race/ethnicity)**
- **Educational background and prior experience with international education**
- **Initial perceptions or expectations about instructors**



Joystick Experiment

- **19-minute lecture about international education**
- **participants asked to change the joystick feely as they are listening to the lecture to their understanding**
- four indicators

+ Good instruction<->Bad instruction

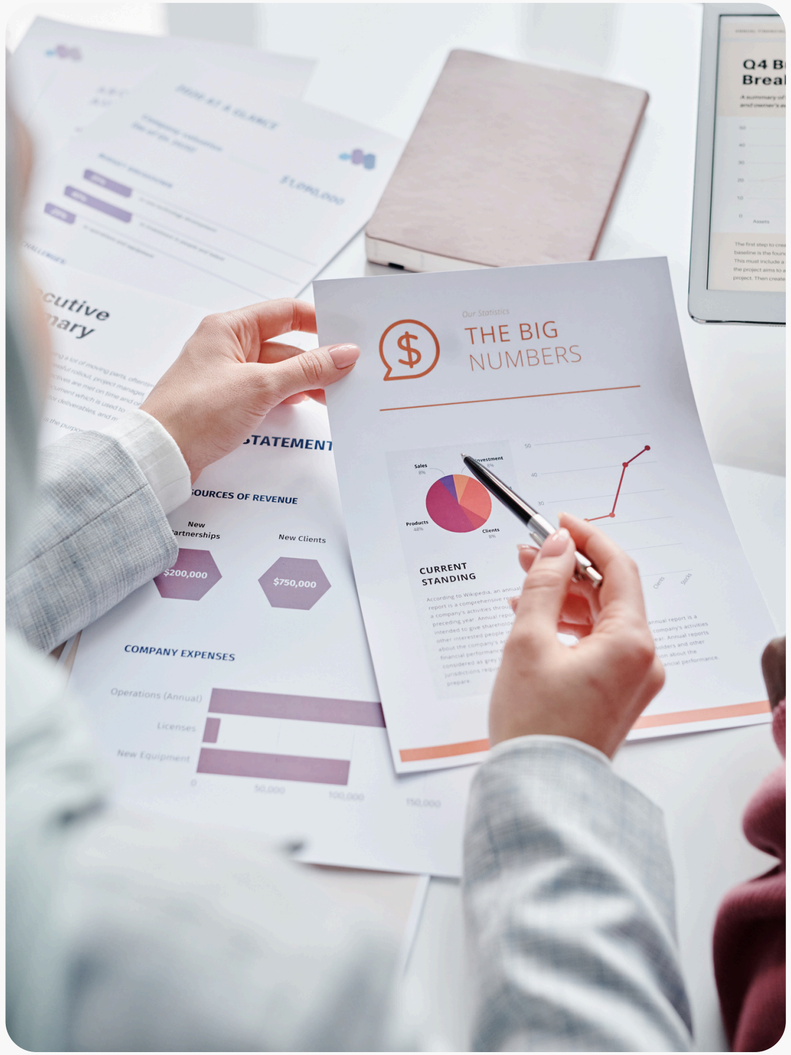
+I am learning/engaged <->not learning/engaged

- Joystick records every 2 seconds

Post questioniares
& Debrief

- **Evaluations of instructor** (e.g., clarity, organization, relatability)
- **Reflection on personal engagement and learning**

Statistical Analysis



Demographics

- **Undergraduate psychology students (mostly freshmen)**
- **Need more info**

Statistical Analysis

- We used a **t-test to see if there was a significant difference** between the two instructors in the two ratings.
- By comparing the means of the data, **we concluded that the male instructor did better in both criteria.**

Data Analysis

Evaluation of Instructor

	Mean	SD	t	Sig
Male	17.20	29.18	**16.69**	**< 2.2e-16**
Female	7.08	5.45		

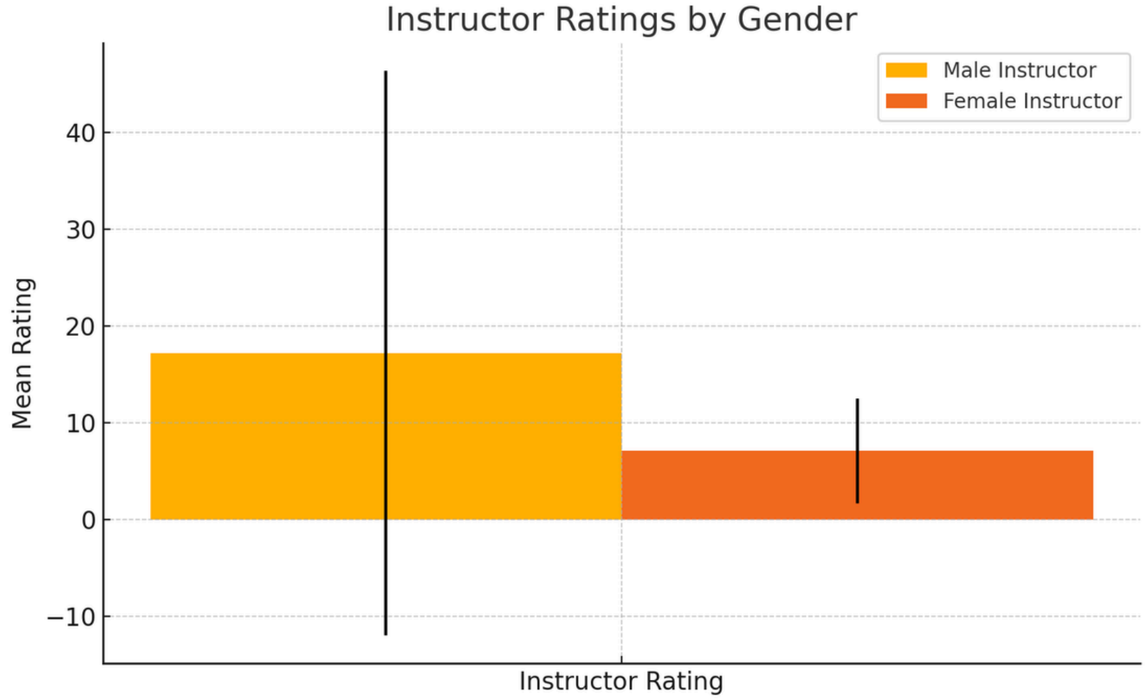
Level of Engagement

	Mean	SD	t	Sig
Male	67.30	43.46	**69.58**	**< 2.2e-16**
Female	14.63	12.84		

Data Analysis

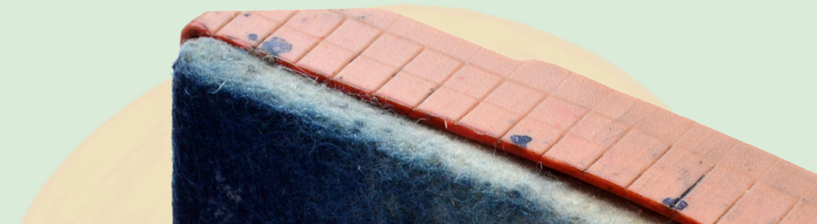
Result 1

- **Male instructors received significantly higher ratings**
- **Large variability in male instructor ratings**
- **T-test result: t = 16.69, p < 2.2e-16 → statistically significant difference.**



Connect to literature

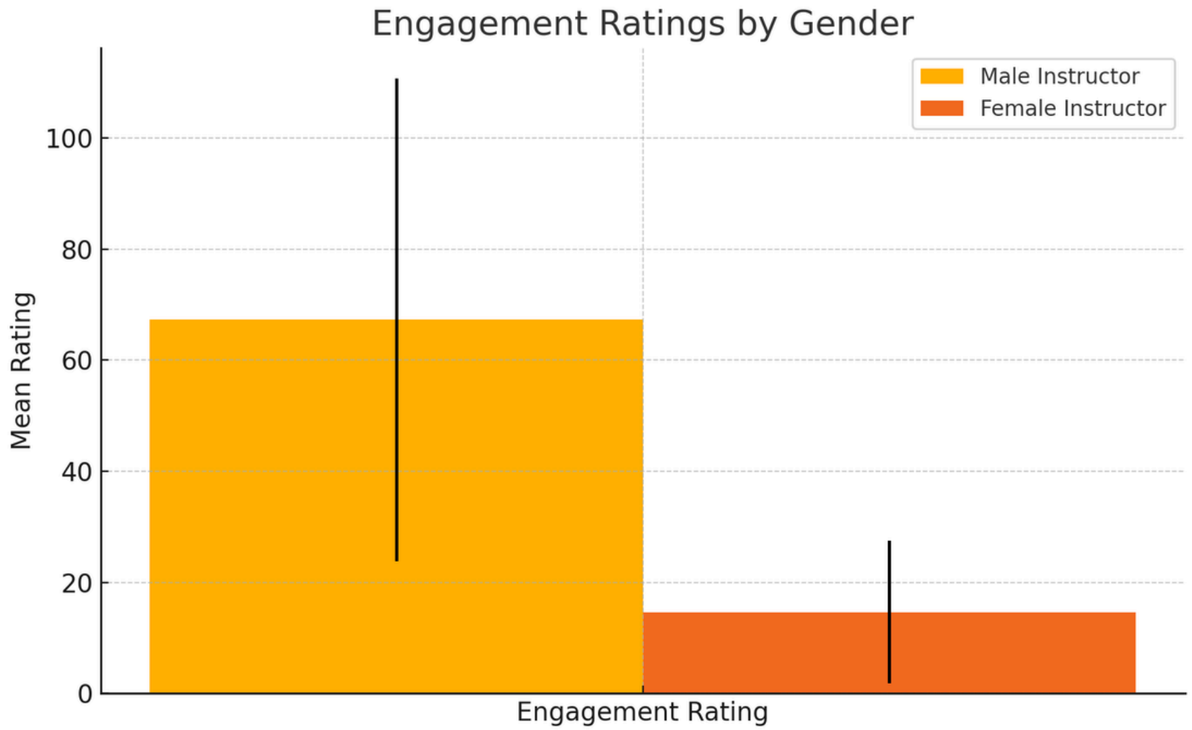
- **MacNell et al. (2015)**
  - “Students rated the male identity significantly higher than the female identity, regardless of the instructor’s actual gender, demonstrating gender bias.”



Data Analysis

Result 2

- **Students rated male instructors much higher on engagement.**
- **Male instructor engagement ratings showed higher variability.**
- **T-test result: t = 69.58, p < 2.2e-16 → highly significant difference.**

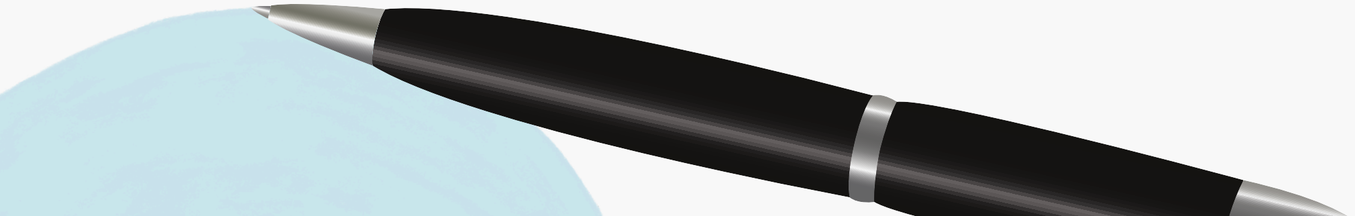


Connect to literature

- **Hoorens et al., 2021**
  - “Students expected to learn more from male than from female professors.”

Limitations/Considerations

1. **Data collection from different years**
2. **Intersecting identities** woman, and Asian
3. **Participants are mainly intro psych students.**
4. **Flag Participants not being able to concentrate** during the data collection
5. The **content of the lecture might also be a confounding factor**
6. **Familiarity with online lectures might be different before and after COVID-19 online education exposure**



Conclusion

In the education systems, there could be a bias towards male professors who are seen as more capable than female-identifying professors. Additionally, those with intersecting identities may be viewed as less engaging and less capable. However, we do not know exactly which factor showed the strongest influence. To assess this, we need a more in-depth experiment.

Thank you